

SUHYEON YU

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EDUCATION

Rice University Master of Computer Science (GPA: 3.91/4.00), Korean Graduate Student Association	Houston, TX, USA Aug. 2025 – Dec. 2026
Sungkyunkwan University B.S. in Computer Science and Engineering	Suwon, South Korea Mar. 2018 – Aug. 2024

EXPERIENCE

Software Engineering Intern <i>ABC Fitness</i>	Jun. 2026 – Aug. 2026 Austin, TX, USA
Frontend Modernization: Rewrote the user profile page from scratch in React, replacing jQuery as part of a company-wide modernization initiative for ABC Trainerize, which serves 400,000+ fitness professionals and 1.6M+ clients worldwide. Async API Coordination: Implemented asynchronous workflows with RxJS that coordinated multi-endpoint profile API requests, suppressed stale responses, and prevented overlapping operations across profile editing and uploading. Unit Testing: Wrote Jest unit tests to verify permission- and role-based access control, conditional rendering, and prevention of overlapping requests during photo/video uploads and profile updates, achieving 97% branch coverage of profile business logic. UI/UX Redesign: Redesigned the user profile interface beyond the original migration scope, improving information hierarchy through a streamlined profile header, grouped content cards, and modal-based profile editing. Cross-Functional Collaboration: Collaborated with a UX designer to refine UI details and a QA analyst to verify functional correctness, delivering UX improvements such as drag-and-drop certification reordering to help trainers highlight their top credentials. AI-Assisted Development: Accelerated development by orchestrating multiple AI agents to build profile sections in parallel in separate Git worktrees and streamlined the development workflow using Figma MCP and team-shared agent skills. Incremental Delivery: Decomposed the profile rewrite into independently reviewable stacked PRs and used a LaunchDarkly feature flag for gradual rollout.	

PROJECTS

Machine Unlearning Comparator <i>React, TypeScript, D3, Zustand</i>	
AI Interpretability: Built the frontend end-to-end for an interactive visualization dashboard for evaluating how machine unlearning affects the accuracy, efficiency, and privacy of ML models, earning 100+ GitHub stars. Data-intensive Visualization: Implemented D3/Recharts visualizations, including 2,000-point UMAP scatterplots of penultimate-layer activations and 400-point stacked dot plots that simulate membership attack inference attacks via a draggable threshold. Interaction Engineering: Optimized hover, zoom, and drag interactions by replacing full SVG redraws with D3 joins and in-place updates, cutting per-hover DOM manipulations by 99%. Backend Extension: Extended the FastAPI/PyTorch backend to a face-recognition dataset, enabling another research lab to adopt it.	
PyTorch-CKA <i>Python, PyTorch</i>	
AI Explainability: Built an open-source Python library that compares internal representations across layers of CNNs and LLMs, addressing slow computation and memory buildup in existing CKA tools. Performance Optimization: Reduced HSIC trace complexity from $O(B^3)$ to $O(B^2)$ per layer pair (B: batch size) by replacing full matrix multiplication with Frobenius inner products, achieving an overall 5.7× speedup over an existing library on H100. Memory Optimization: Implemented automatic activation cleanup and forward-hook removal after CKA computation, reducing retained GPU memory by 95.7% compared to an existing library (ResNet-18, batch size 1,000). Deployment & Maintenance: Published on PyPI (~400 monthly downloads) and automated CI testing across Python 3.10–3.14.	
Online MyeongNyunDang <i>Next.js, TypeScript, TanStack Query, Redux, Tailwind CSS, Docker, AWS</i>	
Frontend Development: Built Next.js frontend features for an education platform serving 5,000+ students, including an admin console integrating REST APIs for course management, homepage curation, and announcement publishing. Initial Load Time Optimization: Cut homepage initial load time by ~40% (5 to 3 sec) using server-side rendering and image lazy loading; implemented on-demand static generation for announcement detail pages.	

PUBLICATION

J. Lee, S. Yu, Y. Jang, S. S. Woo, J. Jo. *Unlearning Comparator: A Visual Analytics System for Comparative Evaluation of Machine Unlearning Methods*. *IEEE Transactions on Visualization and Computer Graphics*, vol. 32, no. 3, Mar. 2026.

SKILLS & TOOLS

Languages: TypeScript, JavaScript, Python | **Frontend:** HTML, CSS, React, Next.js, Redux, TanStack Query, Tailwind CSS, SCSS, D3, Zustand
Backend: Node.js, FastAPI, GraphQL, REST APIs | **Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB) | **Testing:** Vitest, Jest, pytest
Cloud & DevOps: AWS (EC2, S3, Lambda), CI/CD (GitHub Actions), Docker, Git, GitHub | **AI Tools:** Cursor, Claude Code, Codex, Grok Build